



Team members' names:

1. Josh Bae
2. Nikolas Cullifer
3. Tessa Hunt
4. Yash Mahadik
5. Mouni Surapaneni
6. Ritika Venketesh

Semester Project Planning Document

Purpose of Analysis Section

a. What is the problem you want to study (i.e., your curiosity)?

We intend to assess how Virginia Tech students experience safety on and around the Blacksburg campus by focusing on (1) where and what types of safety risks occur, (2) how these experiences vary across student groups, (3) what locations and types of crimes are taking place on and around campus, and (4) which interventions stakeholders should prioritize to maximize perceived and real safety.

b. What questions would you need to answer to provide information that would help solve the problem you stated in a.?

We would need to answer the following questions to provide information for solving our core problem:

1. How do safety perceptions differ across gender, academic role, and disability status, and which groups feel most at risk by time of day?
2. Are students aware of key safety resources, and does familiarity with those resources correlate with confidence in incident resolution?
3. How do primary modes of transportation shape safety experiences, and which commute methods carry the greatest perceived risk?
4. Where are crimes most geographically concentrated, and do those hotspots align with where students feel least safe?
5. When do crimes most commonly occur by time of day and period of the year, and do those patterns match student perceptions of risk?
6. How are reported crimes ultimately resolved, and do resolution patterns vary across offense types?
7. Where do student perceptions of safety and confidence in reporting align with or diverge from ground reality as captured in the VTPD crime records, and what does that gap reveal for stakeholders?



c. What answers would you need to give your audience(s) to help them make decisions to solve the problem in a.?

Based on the problem statement highlighted in (a), and the corresponding questions highlighted in (b), we need to give the following answers to our audience(s) to help them make relevant decisions:

1. An understanding of which groups of students feel least safe, and whether certain populations experience these issues more frequently than others.
2. Insight into whether student perceptions of safety align with actual reported incidents from the VTPD crime logs.
3. An assessment of how aware students are of existing campus safety resources and policies, and whether greater awareness is associated with stronger perceptions of safety.
4. A prioritized list of the most significant safety issues based on both frequency and severity.
5. Actionable recommendations for stakeholders on where resources, policy changes, or interventions should be focused to improve safety and accessibility on and around the Virginia Tech campus.



Write down the justification/motivation for your analysis (i.e., the who and the what).

a. Who wants to know the answers to your analysis (your audience)?

Our primary audiences are as follows:

1. DABS (Dean's Advisory Board of Students)
2. Student Org Coalition (SOC)
3. Services for Students with Disabilities (SSD)
4. VT Facilities
5. Blacksburg Transit (BT)

These organizations can use our analysis to inform their decisions concerning campus safety priorities, communication, transportation, and environmental improvements on the Virginia Tech campus.

b. What is it you need your audience to know and do?

Our audience should understand the main safety challenges students are currently facing, where those challenges are concentrated, and which student groups are most affected. They should also understand how student perceptions compare with reported incident patterns and existing safety resources described in the Clery Act and VTPD documents. Incorporating these insights would help them distinguish between high-concern areas that might be underreported and areas with both high concern and documented incidents.

They can use this information to prioritize five targeted interventions that directly respond to the dashboard findings, assign clear owners such as VTPD and Housing for Pritchard Hall, VTPD and Student Affairs for case outcome communication, Blacksburg Transit for evening route review, Facilities with DABS/SOC for nighttime walking audits, and Student Affairs for orientation communication, and define success measures. This would ultimately enable coordinated action across transportation, facilities, student affairs, campus safety, and student leadership to address the specific issues and locations that matter most to students.

c. Practical implications – how will people be able to use your analysis?

The stakeholders identified in (a) can use these findings to guide campaigns and make more student-centered decisions about campus infrastructure, transportation, and safety priorities. The results can point to specific problem areas students repeatedly experience, helping decision-makers focus on fixes that will have the most real-world impact rather than guessing what students need.

Specifically, stakeholders can identify the problems with the greatest impact across different student demographics. They can ultimately translate these findings into a small set of evidence-backed actions: a Pritchard Hall safety review, quarterly case outcome communication, an evening BT reliability review, a gender-informed walking audit, and an orientation safety brief, and assign an owner and timeline to each. To ensure that these actions yield significant improvements in overall on-campus safety, the dashboard could be used to track key performance indicators, such as nighttime safety perceptions, confidence in reporting, case-resolution awareness, transportation-disruption frequency, and location-specific incident trends.



The ultimate goal of our analysis is to ensure that it is easy for the decision-makers to move from 'general concerns' to specific locations, affected groups, and eventually implement measurable steps by reducing guesswork and improving accountability.

d. Can you include an anecdotal observation that explains the phenomenon you are examining?

During a recent late-night walk back from the library, the lack of adequate lighting along a stretch of the Drillfield made it difficult to see other pedestrians or identify those nearby. Several students mentioned taking longer and better-lit routes instead, even when those paths added significant time to their commute. This small but telling behavior reflects a broader pattern: when students do not feel safe on the most direct routes, they adapt in ways that are inconvenient and sometimes counterproductive. It underscores how gaps in campus infrastructure, whether lighting, visibility, or emergency call station placement, can quietly erode students' sense of security, particularly at night and in lower-traffic areas.



Goals of your Analysis Section

a. What actionable outcomes do you hope to achieve through your analysis?

Our analysis aims to produce a decision-ready set of Tableau dashboards that (1) locate the most common safety challenges, (2) show how experiences differ across student groups and travel modes, and (3) recommend high-impact interventions.

Some of the actionable outcomes include the following:

1. A ranked list of top problem locations or situations, including Pritchard Hall as the most frequently reported VTPD location
2. Identification of the most-impacted groups and when safety concerns are most pronounced, especially after dark
3. A short list of prioritized recommendations with suggested owners and KPIs
4. A baseline set of metrics that can be reused for future surveys to track progress over time
5. A communication-focused finding showing that students' confidence in reporting is lower than the VTPD case outcome data might suggest

b. What types of findings do you think you will need to convince others of the validity of your findings, the interpretation of the findings, and your suggested actionable outcomes?

To convince others of the validity of our findings, we need evidence that is internally consistent, robust across reasonable analytic choices, and triangulated with external sources. With 114 survey responses collected, we have a working dataset, though representation across all key groups is uneven. We will practice transparency by reporting subgroup sizes explicitly, flagging where groups are too small to support strong conclusions, and treating results from those groups as directional rather than definitive.

The VTPD Monthly Crime Logs and the Virginia Tech Annual Security Report serve as important external validators. Where survey-reported safety concerns align with documented incident patterns, that convergence strengthens the credibility of our findings. Where perceptions diverge from the official record, we will note the gap and discuss potential explanations such as underreporting or heightened concern in low-incident areas.



Objectives of Analysis (Methods)

a. What data do you need to perform your analysis and create your data story?

Our analysis would draw on three datasets. The primary source is our Safety and Accessibility survey, covering student demographics (Gender, Role, Disability Status, Mode of Commute), perceived safety by location and time of day, resource awareness, and transportation barriers. This is supplemented by two institutional resources, which are highlighted below:

1. The VTPD Monthly Crime Logs from January 2022 to March 2026, which provide time and location-stamped incident records, will help us compare student-perceived risk areas against documented incident patterns.
2. The Virginia Tech 2024 Annual Security Report, which documents three-year crime trends and institutional safety policies, would help us contextualize our survey findings within a broader historical and policy framework.

b. What sources did you use to guide your data collection?

We primarily utilized personal experience and anecdotes from other students, faculty, and staff to formulate questions relevant to our topic of safety. We then cross-referenced those themes against the VTPD crime log and the Clery Act report to ensure our questions reflected documented incident patterns. For example, asking specifically about late-night safety and residential area concerns, given the concentration of alcohol-related incidents in on-campus housing.

c. What steps did you take to integrate your data?

We exported the Google Form responses as a CSV file and imported them into Google Spreadsheets for initial review. Furthermore, we extracted the VTPD crime logs from their respective PDFs and converted them into a structured tabular format to facilitate further analysis. Specifically, to enable map-based visualizations in Tableau, we enriched the VTPD crime log with geographic coordinates by matching location strings to a reference table of campus building coordinates, assigning latitude and longitude values to each incident record.

From the Clery Act report, we'll specifically use data on VT's safety workshops conducted throughout the year (2024 Violence/Sexual Violence Prevention Programs) and on reports of each type of offense reported on- and off-campus, from 2022 to 2024. We'll use data specific to the Blacksburg campus, as that is the primary focus of our analysis. As with the VTPD crime log, we'll need to convert the data from PDF to a spreadsheet for easier analysis.



d. What steps did you take to clean your data?

A. Cleaning the Survey Data:

Most survey questions used controlled response options (For example, single-select and multi-select), which eliminated the need for extensive manual cleaning. For one of the multi-select fields, we applied regex patterns in the original spreadsheet to standardize delimiters.

This step was crucial, as some options in our multi-select used the same delimiter as the spreadsheet software used to separate two different responses. Once the delimiter was modified, we separated the appropriate responses into different columns. For ease of analysis in Tableau, we pivoted these newly created columns to extract a column that would ultimately yield an accurate count of respective entries. We also converted the ordinal scales from text to numerical values.

Finally, the one optional free-response question requires additional cleaning, with a specific focus on removing unwanted or inappropriate content and correcting spelling inconsistencies.

B. Cleaning the VTPD Crime Logs:

Stage 1: Geocoding

The first challenge was spatial. Each incident record contained a raw address string, but Tableau's mapping features require structured latitude and longitude coordinates. The pipeline initially attempted geocoding through the Census Bureau's geocoder, but it proved unreliable for the kinds of addresses in the dataset, many of which were campus building names, partial addresses, or non-standard strings. The switch was made to the Google Maps API, which handled ambiguous addresses significantly better.

Even with Google Maps, a small subset of addresses remained unresolvable, either because they were too vague, used internal campus naming conventions, or simply weren't recognized. For these, a **MANUAL_COORDS** dictionary was maintained: a hardcoded lookup of known problem addresses mapped directly to their correct coordinates. This two-tier approach (API first, manual fallback second) ensured near-complete geographic coverage of the dataset.

Stage 2: Offense Normalization

Raw offense data presented a different kind of complexity. A single incident could involve multiple offenses, and these were stored together in a single cell as a concatenated string rather than separate records. The pipeline used semicolons as delimiters to split these strings into individual offense tokens.

After splitting, deduplication was necessary as some incidents contained repeated offense strings within the same cell. The cleaned offense tokens were then passed to the Claude API for canonical normalization. Because raw offense strings varied widely in phrasing and specificity, Claude mapped each one to a standardized category label, making grouping and filtering in Tableau tractable.



Stage 3: Disposition Standardization

Each incident record also included a disposition field that described how the case was ultimately resolved (e.g., "Closed- Arrest Made," "Referred to Student Conduct," "Inactive"). These raw strings were inconsistent in formatting and phrasing, making aggregation difficult.

An AI agent approach was initially explored for standardization, but it produced inconsistent outputs and was difficult to audit. The pipeline was refactored to use a hardcoded lookup dictionary instead: a manually constructed mapping from each observed raw disposition string to a clean, standardized label. While less automated, this approach was fully deterministic, easy to verify, and produced no unexpected outputs.

Stage 4: Time of Day Derivation

A further derived column, Time of Day, was created from the existing Occurrence Time field. Rather than working with raw timestamps, which are difficult to aggregate and visualize meaningfully, incident times were bucketed into labeled time-of-day categories (Morning, Afternoon, Evening, and Night). This made it straightforward to build time-based charts in Tableau and analyze when different offense types were most likely to occur.

Stage 5: Export

With all transformations applied, the clean dataset was saved in two versions. The first was an original file that preserved the one-row-per-incident structure, in which multi-offense incidents remained together. The second was an exploded file, where each offense was given its own row, making it possible to build offense-level bar charts and filters in Tableau without complex calculations. Both files were loaded into Tableau and used across different sheets and dashboard views.

Stage 6: Offense Explosion via Tableau Prep Builder

With the Python pipeline complete, the cleaned and geocoded dataset was loaded into Tableau Prep Builder to address an issue with the exploded file produced by the Python pipeline. While Python had successfully split and normalized the offense strings, the resulting exploded dataset was producing errors when loaded into Tableau, making it unreliable for building offense-level charts and filters. Rather than troubleshoot the Python output, the explosion step was rebuilt directly in Prep Builder, where the transformation could be applied to the clean base file in a more controlled environment.

The flow begins by applying a Quick Calc to the Offense field, collapsing any runs of extra whitespace and trimming the edges of the string. From this shared cleaning step, the flow branches into two paths. The first branch writes the dataset directly to a Hyper output file, preserving the original one-row-per-incident structure for dashboards where the unit of analysis is the incident itself.

The second branch proceeds through a splitting and unpivoting sequence. Seven calculated columns are added in sequence using Prep Builder's SPLIT() function, each extracting a different comma-separated token from the Offense field. The seventh column receives an additional Quick Calc to normalize any internal spacing.



An Unpivot step then rotates these seven columns into rows, producing two new fields: one holding the offense value and one recording the source column position. Both generated columns are renamed immediately after the unpivot for clarity.

A final cleaning container removes the null rows produced by incidents with fewer than seven offenses, trims each remaining value once more, and drops both the intermediate working column and the original concatenated Offense field. The cleaned offense value is retained in a column named Offenses (E), and the resulting one-row-per-offense dataset is written to a second Hyper output file. Both files were loaded into Tableau and used selectively across dashboard sheets depending on whether the analysis required incident-level or offense-level granularity.

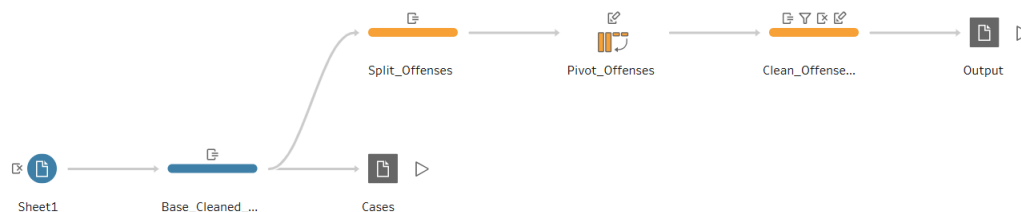


Figure 01: Tableau Prep Workflow for the VTPD Dataset

e. Describe your statistical analysis. What did you do, how did you organize it, and what did your testing reveal?

With 114 survey responses collected, our analysis relied entirely on descriptive statistics. For each survey item, we computed frequency distributions and proportions to identify the most commonly reported safety concerns, gaps in resource awareness, and transportation-related risks. For the VTPD crime log, we summarized incident counts by offense type, location category, time of day, and case outcome.

We organized our findings across seven Tableau dashboards, each addressing a distinct analytical question. The first dashboard segments survey responses by student demographics to surface differences in how various student populations experience campus safety. The second dashboard examines transportation safety and commute-related perceptions. The third dashboard examines resource awareness and confidence in incident resolution. The next three dashboards draw primarily from the VTPD crime log, presenting the geographic distribution of incidents, their temporal patterns across time of day and year, and the outcomes of reported cases. The final dashboard compares student perceptions with documented incident patterns and reporting outcomes.

Our findings revealed that women reported lower perceived safety than men, particularly at night. Awareness of campus safety resources was uneven across academic roles, with freshmen showing the highest confidence in reporting. Geographically, incidents were concentrated in residential and parking areas, and most crimes occurred during evening and nighttime hours. Larceny/theft, vandalism, and alcohol-related offenses were among the most frequently reported offense types. The final comparison dashboard also showed that several top offenses had very high resolution rates, often through arrest or referral to Student Conduct, which helps explain why the gap between student confidence and official case outcomes is an important communication opportunity. Because several subgroup samples are small, we treat cross-group comparisons as directional rather than statistically definitive.



f. What artifacts from your analysis are important to carry forward to tell your story?

Some potential artifacts that might be important to carry forward to tell our story are as follows:

1. Frequency distributions of the most commonly reported safety concerns
2. Subgroup comparison charts showing how perceptions differ across gender, disability status, and commute mode
3. A side-by-side view of student-perceived risk locations and VTPD-documented incident locations
4. A resource awareness summary showing familiarity with campus safety tools and whether awareness correlates with higher safety ratings
5. Prioritized recommendation table highlighting key owners and KPIs



A reference section detailing all sources you used.

a. Where did you get your data (provide titles of the pages and links)?

Sr. No.	Data Source	Link
1	Group Survey Results	[Attached to the Checkpoint Submission]
2	Virginia Polytechnic Institute and State University – 2025 Jeanne Clery Campus Safety Act Report	https://police.vt.edu/content/dam/police_vt_edu/clery-reports/2024%20ASR.pdf
3	VTPD Daily Crime Logs – January 2022 to March 2026	https://police.vt.edu/crime-stats/crime-logs.html

b. What references did you use to plan your project? For example, what news articles or magazine articles did you use to help analyze or interpret your data (provide titles of the articles and links)?

We have referred to the following references to help with data collection and getting a concrete understanding of our target domain:

Sr. No.	References	Link
1	Virginia Polytechnic Institute and State University – 2025 Jeanne Clery Campus Safety Act Report	https://police.vt.edu/content/dam/police_vt_edu/clery-reports/2024%20ASR.pdf
2	VTPD Daily Crime Logs – January 2022 to March 2026	https://police.vt.edu/crime-stats/crime-logs.html
3	Virginia Tech Accessibility Portal	https://accessibility.vt.edu/

c. What data visualizations did you use for inspiration (provide the links for the visualizations)?

Our dashboard design was informed by Tableau examples reviewed during coursework (Labs and In-Class Discussions), which demonstrated the effective use of bar charts and treemaps to communicate distributions across demographic subgroups. These examples also guided broader design principles that we implemented in our dashboards: a cohesive and complementary color scheme to distinguish gender groups across charts, transparency about any exclusions through chart titles, openness about data sources and authorship through a dedicated footnote panel, and the removal of redundant axis ticks where data labels already communicate the values clearly. The following page contains a screenshot of our dashboards.

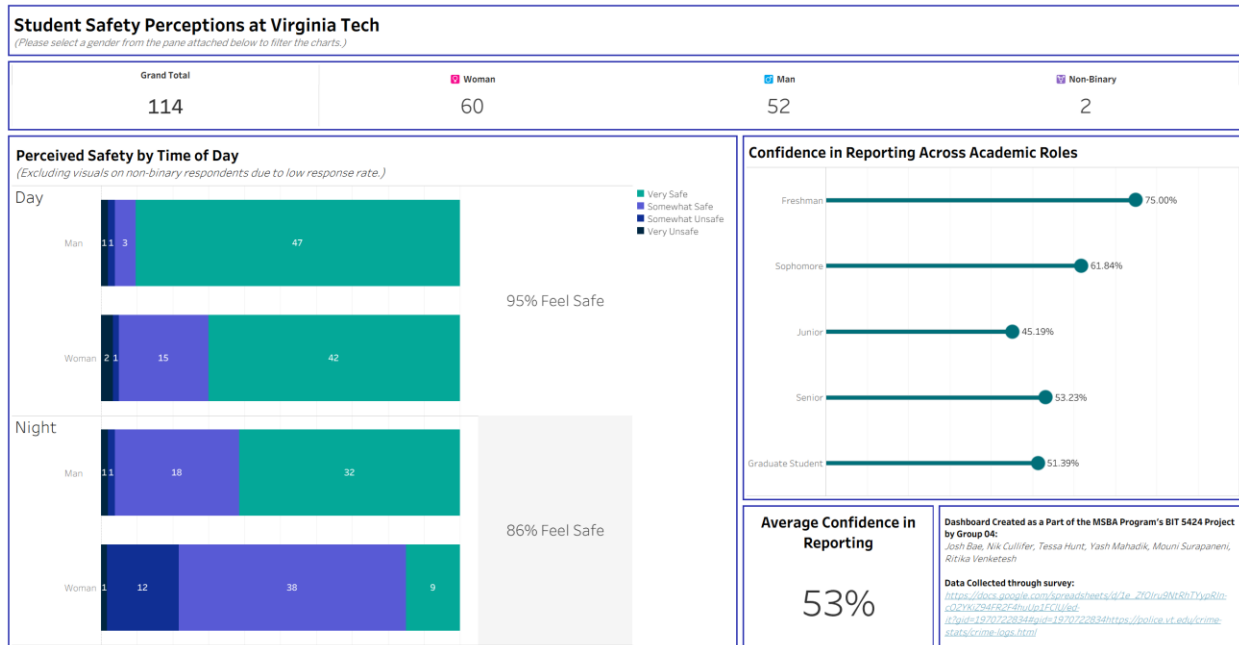


Figure 02: Dashboard Portraying the Differences in Safety Perceptions for Varying Student Demographics

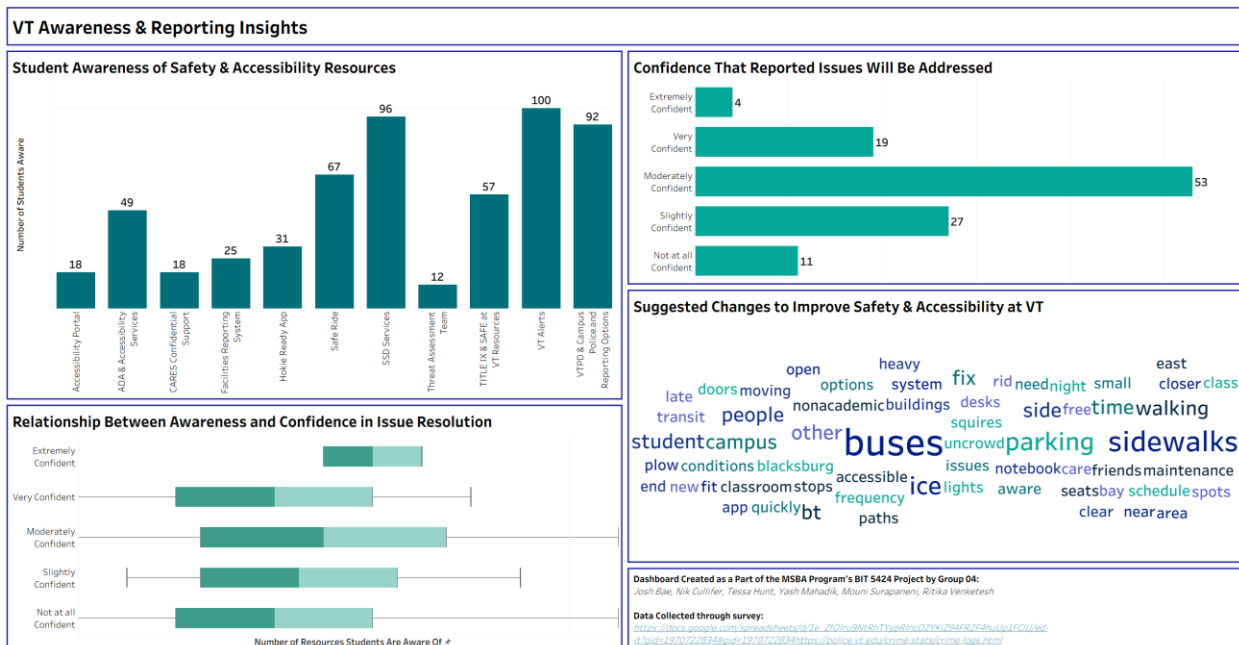


Figure 03: Dashboard Portraying the Awareness about Key Safety Resources and Confidence in the Resolution Processes among Students

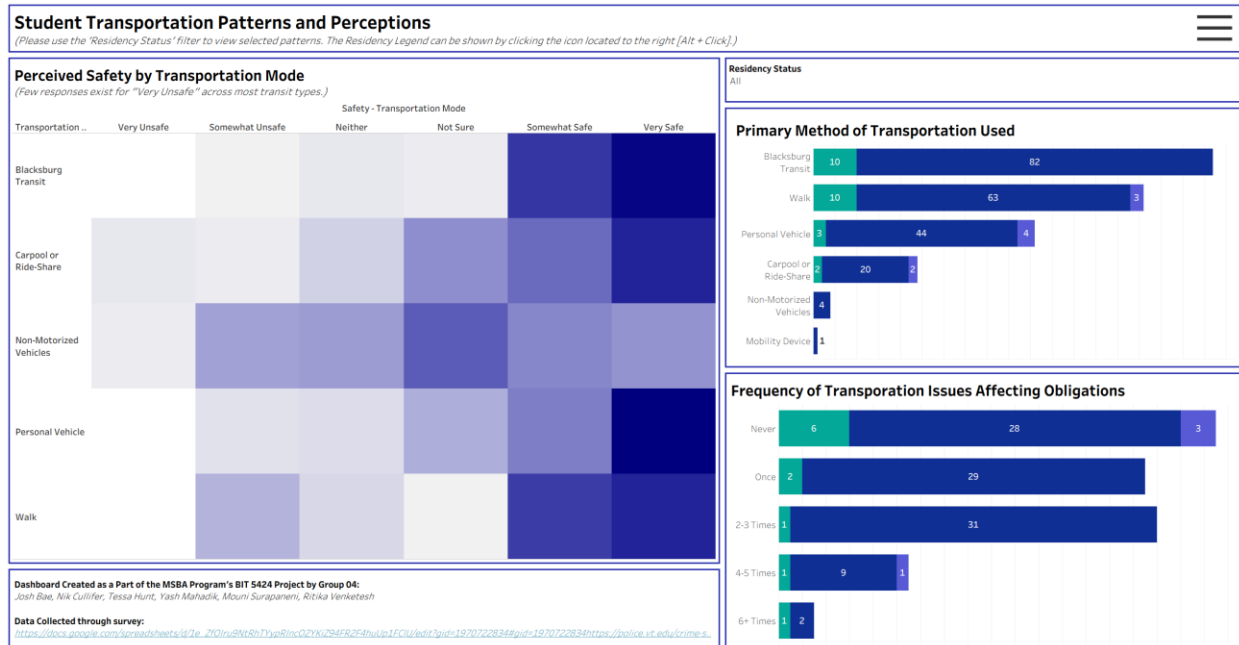


Figure 04: Dashboard Portraying the Safety Perceptions Based on Transportation Preferences

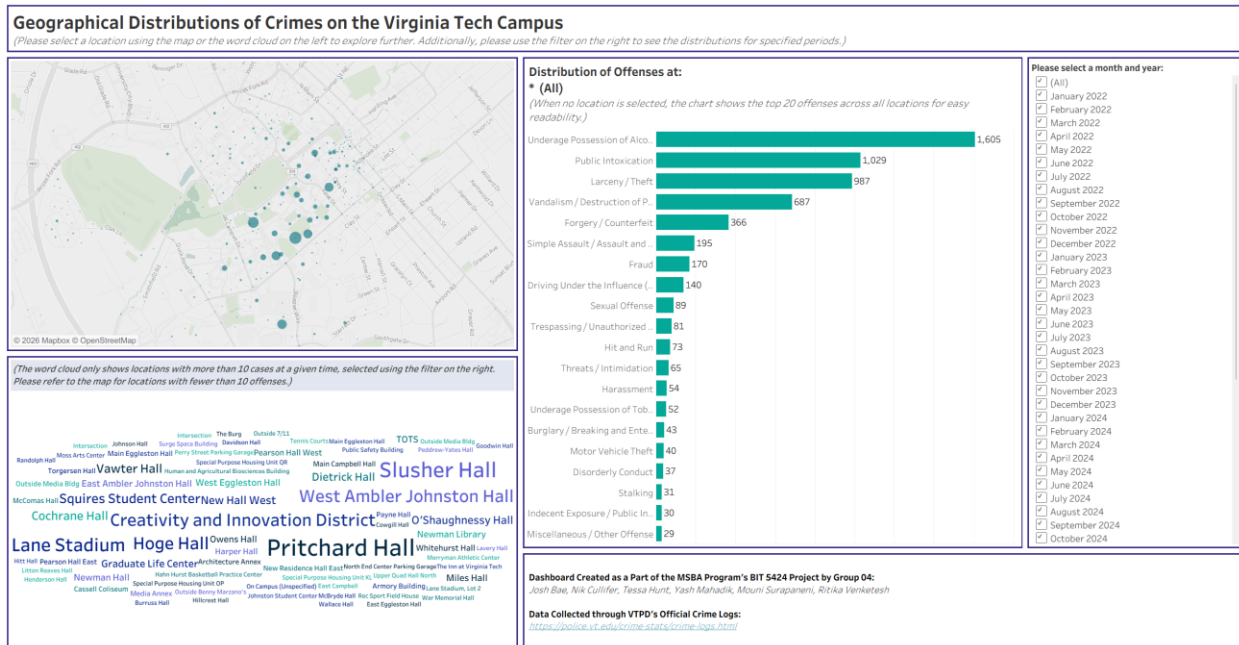


Figure 05: Dashboard Portraying the Geographical Distribution of Crimes on the Virginia Tech Campus



Pamplin College of Business Business Visualization Analytics/ Spring 2026

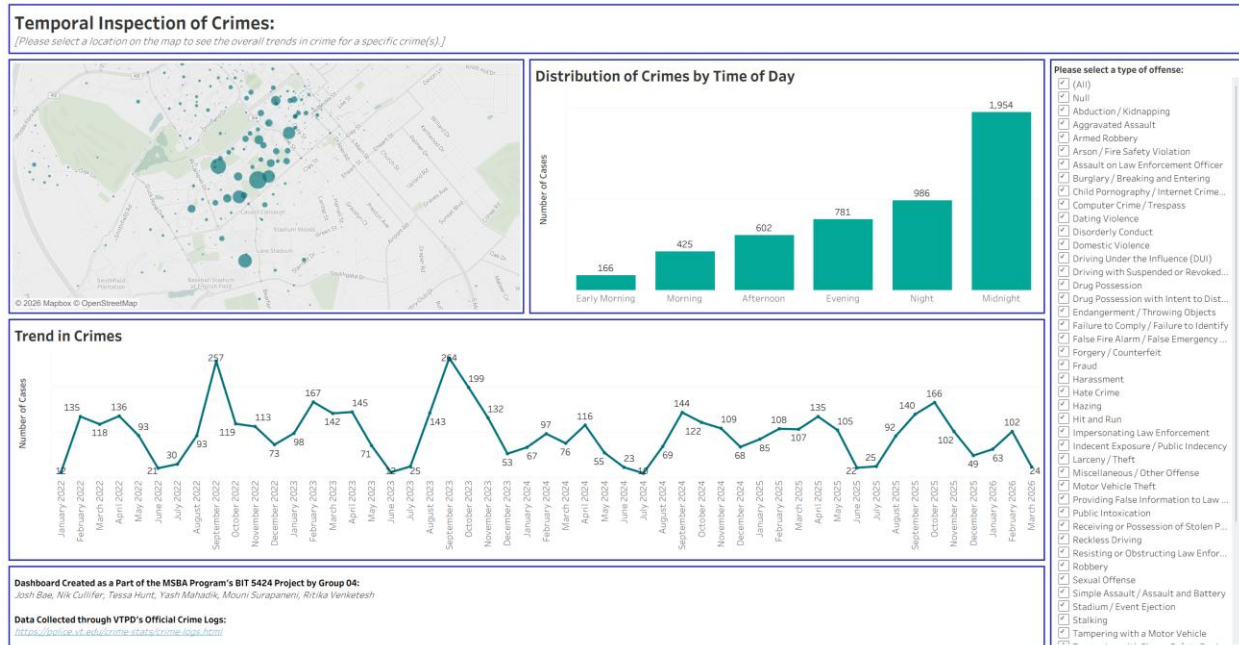


Figure 06: Dashboard Portraying the Temporal Distribution of Crimes on the Virginia Tech Campus

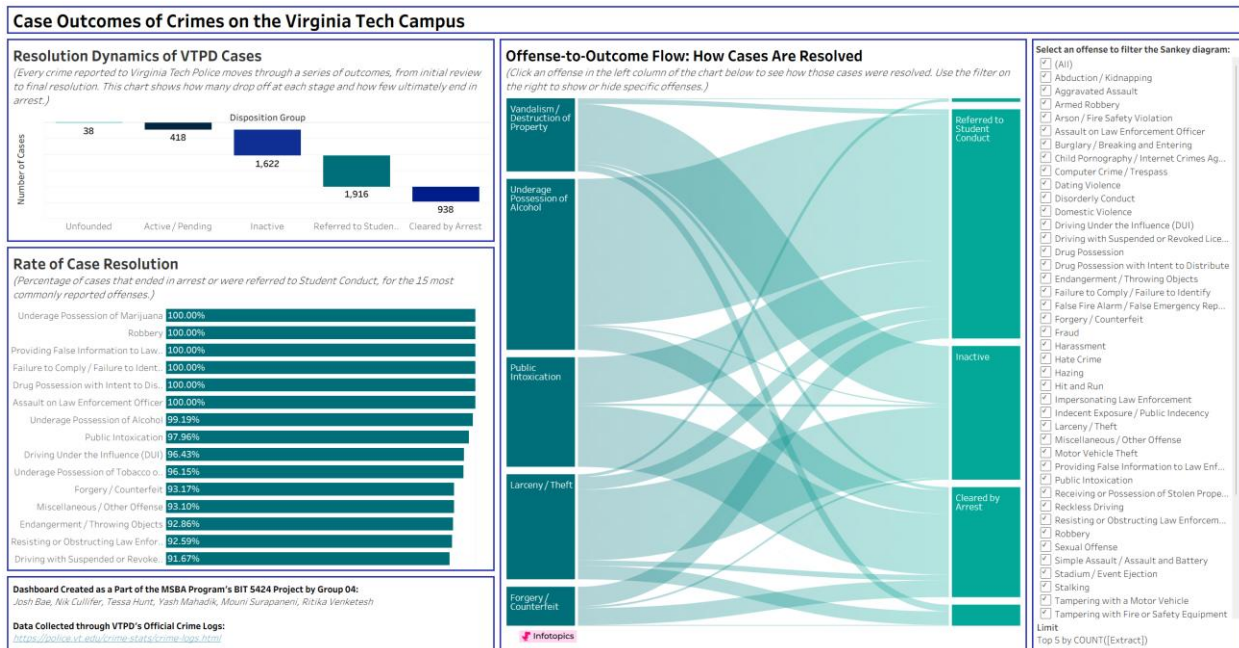


Figure 07: Dashboard Portraying the Case Outcomes of the Crimes Reported on the Virginia Tech Campus

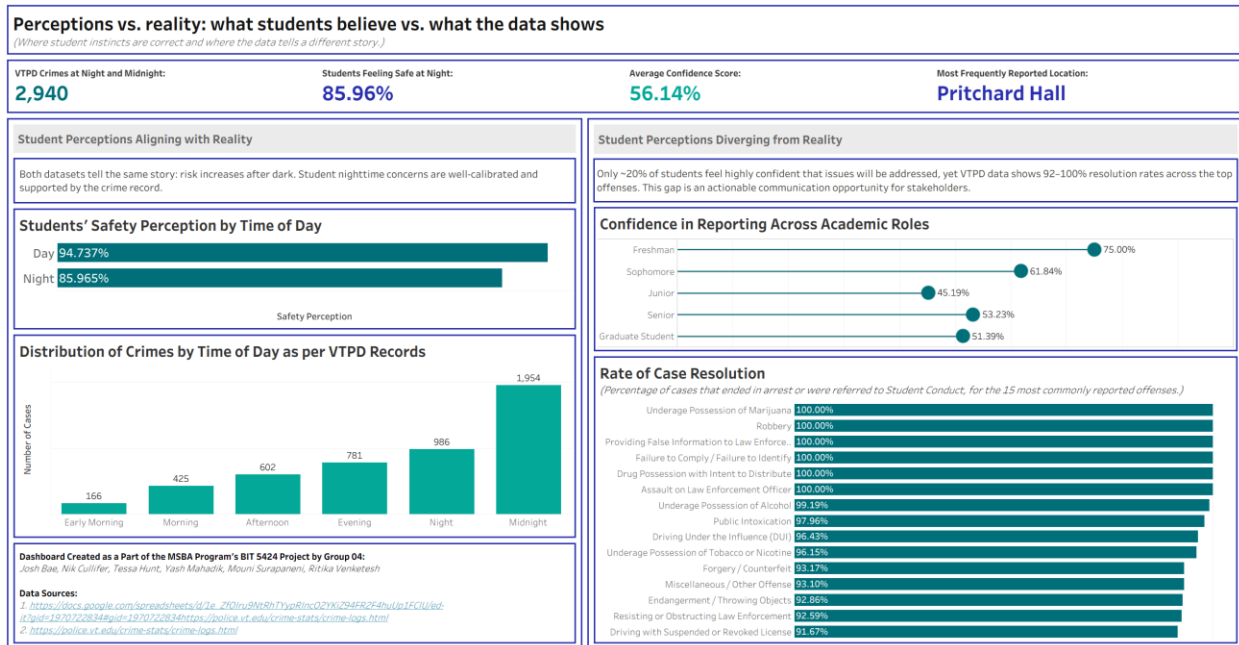


Figure 08: Dashboard Concluding the Alignments and Diversions between Our Collected Datapoints and VTPD Data

- d. What references did you use to help interpret, explain, or provide recommendations for your findings (provide titles and links)?

The Virginia Tech 2024 Annual Security Report and the VTPD Monthly Crime Log will serve as key references for contextualizing our findings, as they document institutional safety trends and incident patterns directly related to the safety concerns raised in our survey.